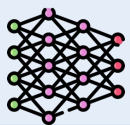


Input

User History s



Sequence Encoder h_ϕ
(Transformer)



History Representation

e_s



Target Item y



Item Representation

e_y

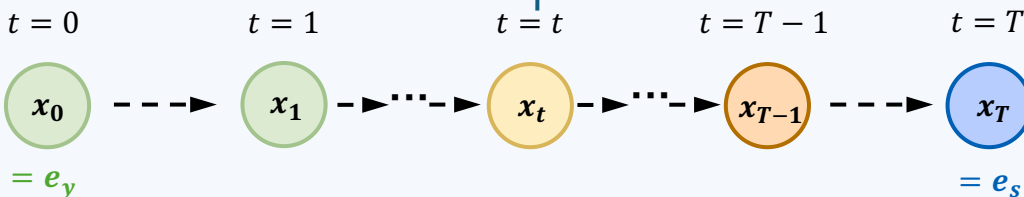


Brownian Bridge Architecture

Forward Process (Item $\rightarrow$ History)

$$q(x_t|x_{t-1}, x_T = e_s)$$

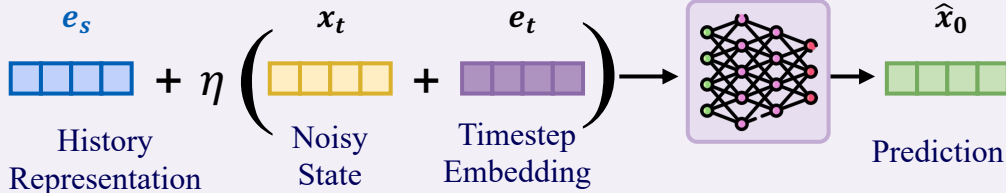
$$\text{Noise Variance } \delta_t = m \cdot \beta_t(1 - \beta_t)$$



Reverse Process (History $\rightarrow$ Item)

$$p_\theta(x_{t-1}|x_t, x_T = e_s)$$

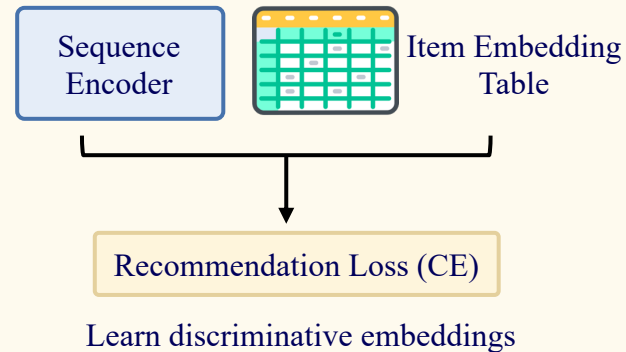
Denoiser g_θ



- x_0 Target Item
- x_t Intermediate State
- x_T User History
- $\rightarrow$ Forward (noising)
- $\leftarrow$ Reverse (denoising)

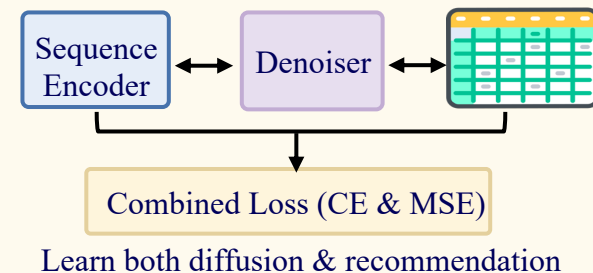
Progressive Learning Strategy

Stage 1: Sparse Embedding Pretraining



Stage 2: Preference-Centric Diffusion Training

- Load Stage-1 embeddings
- Jointly optimize all components



Inference (History $\rightarrow$ Item)

